

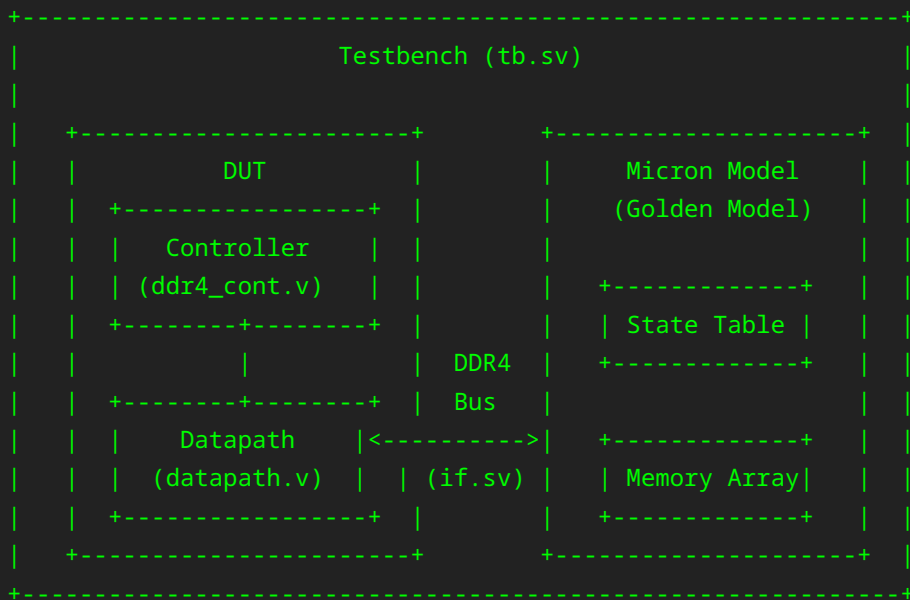
DDR4 Controller Verification Environment

Micron Golden Model Integration & Test Suite

1. Overview

This project implements a robust **DDR4 Memory Controller** and **Datapath**, verified against the industry-standard **Micron DDR4 Behavioral Model**. The setup focuses on protocol compliance, timing accuracy, and reliable initialization sequences for DDR4 SDRAM devices.

2. System Architecture



- **Controller:** Manages JEDEC state machine (Initialization, Refresh, Row Management).

- **Datapath:** Handles DQ/DQS strobe alignment, gear-shifting, and read/write data buffering.
- **Micron Model:** Acts as the "Golden Reference" to validate that the controller adheres to DDR4 timing and protocol specs.

3. File Structure & Components

Core Verification Files

- `tb.sv` : Top-level SystemVerilog testbench. Instantiates the DUT and Golden Model.
- `interface.sv` : Defines the `DDR4_if` containing all physical layer signals (DQ, DQS, ACT_n, etc.).
- `subtest.vh` : Contains the test sequences, including power-up, MRS configuration, and R/W operations.
- `timing_tasks.sv` : Helper tasks for verifying JEDEC timing parameters during simulation.

Configuration & Infrastructure

- `arch_package.sv` / `proj_package.sv` : Centralized parameters for memory density (2G-16G), bus width (x4, x8, x16), and speed grades.
- `modelsim.do` : Automation script for compilation and simulation. Defines key macros like `DDR4_2G_X4`.
- `memory_file.txt` : Input file for pre-loading memory contents for specific test scenarios.

4. Verification Methodology

The environment utilizes **Behavioral Verification** through the Micron Model:

- **Protocol Compliance:** The model monitors the command bus (ACT_n, RAS, CAS, WE) and flags any JEDEC state machine violations.

- **Timing Enforcement:** Strict checks for `tRCD`, `tRP`, `tCL`, and other critical DRAM timing parameters.
- **Data Integrity:** Write/Read loopback tests verify that data stored in the `MemoryArray.svp` is correctly retrieved by the Controller.

5. Simulation Guide

How to Run

Ensure you have ModelSim/Quarta installed and all Micron model files (`.svp`) in the working directory.

```
vsim -do modelsim.do
```

Key Configuration Macros

- `DDR4_2G_X4` : Configures a 2Gb density device with a 4-bit bus width.
- `FIXED_1600` : Sets the memory clock frequency to 800MHz (1600 MT/s).
- `MODEL_DEBUG_CMDS` : Enables verbose logging of command transitions in the transcript.

6. Expected Controller Sequence

The controller follows the standard JEDEC power-up sequence:

1. **RESET_n Asserted:** Low duration > 200 μ s.
2. **CKE Transition:** CKE pulled high after RESET_n release.
3. **MRS Commands:** Programming Mode Registers (MR0-MR6) for latency and features.
4. **ZQ Calibration:** Initial ZQCL command to calibrate output drivers.
5. **IDLE State:** Ready to receive ACTIVATE and READ/WRITE commands.